Effects of Visual and Auditory Alert Saliences on Detection Performance During Video-Based Driving Tasks

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Abstract

In vehicle information systems (IVIS), drivers are presented with a continuous stream of visual and auditory alerts that compete with the primary task of driving. The literature contains conflicting findings about how alert modality and alert salience affect detection performance, and the two factors are rarely tested together within a single experimental design. This study examined the effects of alert modality (visual vs. auditory) and alert salience (low vs. high) on detection performance in a video-based driving task. Twenty-four adult participants completed a 2 × 2 within-subjects experiment in which they responded to navigation and notification alerts overlaid on top of prerecorded dashcam footage. Reaction time, miss rate, and response accuracy were all recorded across all four conditions. Repeated-measures ANOVAs revealed a significant main effect of salience on reaction time, $F(1,23)=869.47$, $p<.001$, partial $\eta^2 =.489$, and on miss rate, $F(1,23)= 13.39$, $p< .001$, partial $\eta^2 = .143$, with high salience alerts producing faster reaction times and lower miss rates than low salience alerts in both modalities. The main effect of modality, however, was not significant for any dependent variable. A significant modality × salience interaction was observed for reaction time, $F(1,23) = 39.55$, $p < .001$, with the salience effect larger for visual alerts (166ms) than for auditory alerts (117ms). Response accuracy was at the ceiling across all four conditions tested and could not be used to evaluate the hypotheses. The findings in this study indicate that alert salience is the main contributing factor in detection performance and that visual alerts placed within the driver's central visual field can outperform auditory alerts when sufficiently salient.

Introduction

1.1 Background

Advanced Driver Assistance Systems (ADAS) are now available and standard in many modern vehicles, both warning drivers of hazards and helping reduce collision risk. These systems commonly deliver alerts visually, via dashboard displays or an integrated windshield heads-up display, or audibly via the vehicle's speakers. Though how well they work depends on two things: which sensory modality is used, and how noticeable alert actually is.

Drivers already face high visual demands behind the wheel, monitoring the roadway, reacting to traffic, and reading information from displays, all while competing for the same limited visual resources. Visual In-Vehicle information systems (IVIS) alerts add to that load and can slow both the driver's detection and response. On the other hand, auditory alerts use a separate channel, allowing drivers to receive warnings without taking their eyes off the road, even for a split second.

Ho, Reed, and Spence (2007) found that combined auditory and tactile warnings cut braking reaction time compared to unimodal signals. Relying on a single modality for the driver's primary visual task comes with real costs in response time.

1.2 Problem Statement

The gap in the research is with salience. Most IVIS studies compare modalities without varying the actual alert visibility (Bella & Silvestri, 2017), which limits the usefulness of their findings for real-world design decisions. Little research has examined how salience, meaning how perceptually prominent an alert is, interacts with modality to affect detection and accuracy performance (Einhäuser et al., 2024). This study addresses that gap by manipulating both modality and salience with a 2 × 2 within-subjects experiment.

1.3 Background and Prior Work

Modality and salience have been studied separately for decades, but the bulk of that work has examined only one at a time. Within the modality literature, auditory alerts have shown a reaction time advantage in some studies (Bella & Silvestri; Wang et al., 2022), often attributed to the fact that auditory information does not require the driver to take their eyes off the road. Visual alerts allow more precise spatial information but compete with the driver's existing visual load of the driving environment. Multimodal designs that combine visual and auditory channels have shown additional gains in some contexts, although the magnitude and benefits depend on how the channels are integrated (Biondi et al., 2017; Wickens, 2008)

1.4 Approach of This Study

This study addresses the gap identified above by testing both modality and salience within a single 2 × 2 within-subjects experimental design. Each participant experienced all four combinations of modality (visual vs auditory) and salience (low vs high), allowing direct comparisons across conditions, while controlling for individual differences in baseline reaction time and attention. Detection performance was measured through three dependent variables: reaction time, miss rate, and response accuracy. The within-subjects design and three dependent variables allow for a more complete picture of how the two design factors jointly affect driver performance than either factor would on its own.

1.5 Research Questions

Based on the gap identified in the prior literature, this study examines three distinct research questions:

1. Within the visual modality, does the salience of an alert affect detection performance?
2. Within the auditory alert modality, does the salience of an alert affect detection performance?

3. At high salience, does alert modality affect detection performance?

1.6 Hypotheses

The three paired hypotheses proposed for this study are:

H1 (Visual salience effect): Participants will show a higher response accuracy for high salience visual alerts than for low salience visual alerts.

H0^1 (Visual salience effect): There is no difference in reaction time or response accuracy between high salience and low salience visual alerts

H02 (auditory salience effect): Participants will show a higher response accuracy for high salience auditory alerts than for low salience auditory alerts

H0^2 (auditory salience effect): There is no difference in reaction time or response accuracy between high salience and low salience auditory alerts.

H3 (modality comparison): Participants will show faster reaction times and higher response accuracy for high-salience auditory alerts than for high-salience visual alerts.

H0^3 (modality comparison): There is no difference in reaction time or response accuracy between high salience visual and high salience auditory alerts.

2. Review of Literature

2.1 Multimodal Warning Systems

Earlier IVIS research focused primarily on unimodal alerts, typically visual signals either on the dashboard or instrument panel. Eventually, researchers began asking whether splitting alert information across modalities could improve driver response. Multiple resource theory (Wickens, 2008) claims that humans have separate cognitive resource pools tied to different sensory modalities. That performance drops when two tasks compete for the same pool of resources.

Ho et al. (2007) found that combined auditory and tactile warnings reduced braking reaction time compared to unimodal visual or auditory warnings. Adding a non-visual component to an alert significantly improved the driver's response speed. Drivers already handling a visual task respond faster to warnings that do not demand more of the same resource.

Biondi et al (2017) tested the same idea in two driving simulator experiments with 22 participants in each. Multimodal auditory and tactile warnings produced faster braking reaction times than when either signal was tested independently. The advance held even when drivers were on a hands-free phone call (experiment 1) or driving in dense urban traffic (experiment 2). The auditory-tactile combination worked even in conditions where the drivers were distracted.

Karthaus et al (2020) found some similarities in their study with distractors. In their simulator study, visual distractors slowed braking response times to the lead-vehicle brake light more than auditory distractors did. Visual stimuli compete with the visual primary task in a way that auditory stimuli do not. (This is what multiple resource theory predicts).

These results motivate this study's direct comparison of visual and auditory alert modalities. Where Ho et al. (2007) and Biondi et al. (2017) tested combined models, this study will test each modality separately to measure its individual effects on detection and accuracy.

2.2 Auditory Warning Design Features

Research studies have also examined the properties of auditory warnings in relation to perceived urgency and driver behavior. The foundational work that laid the groundwork for this is Edworthy, Loxley, and Dennis (1991), who conducted eight experiments mapping how the acoustic features of a sound shape their perceived urgency. They found that fundamental frequency harmonic regularity, amplitude envelope, speed, rhythm, pitch range, and atonality each have replicable effects on perceived urgency. Their final experiment showed that warning bursts could be designed to fall along an urgency scale in a predictable order, and listeners agreed on the ranking. The framework has shaped the design of auditory warnings in aviation, medical alerts, and automotive systems for the last three decades.

Baldwin and Lewis (2014) tested the same kind of urgency mapping across modalities. With 20 within-subject participants, they characterized urgency utility functions for visual, auditory, and tactile warnings. Warning parameters within each modality could be mapped onto urgency ratings using Steven's power law. For auditory warnings, urgency was driven mainly by interpulse interval and pulse duration.

Nees, Helbein, and Porter (2016) tested whether speech-based auditory cues improve recall of alerted driving events compared to non-speech tones. Participants who heard speech-based alerts had better recall of the alerts. Alerts that carry verbal information/communication supported both immediate detection and later, better recall. For IVIS design, what an alert communicates may matter as much as whether it plays or not.

Wu, Boyle, Marshall, and O'Brien found a complication. They tested 192 participants on forward collision warning alerts that varied in fundamental frequency and duty cycle. The relationship between acoustic parameters and collision avoidance performance was inconsistent.

The alert rated highest in subjective urgency (high fundamental frequency of 641 Hz) actually produced worse collision-avoidance behavior than moderately urgent alerts. The alert with the lowest duty cycle ("too weak") also underperformed. The best-performing collision-avoidance signal was a low fundamental-frequency tone, even though participants rated it as less urgent than the higher-frequency options on their rating scale. In this study, the most urgent-sounding alert did not produce the best driver response.

Together, these five papers show that the properties of an auditory alert, not just whether one plays, affect how well drivers perform. The relationship between salience and performance within auditory alerts is not always linear. Both findings shape how this study defines auditory salience and how H2 is framed.

2.3 Alert Salience and Detection Performance

Alert salience describes how much a stimulus stands out from its surroundings and draws attention. In IVIS design, a high-salience alert is more noticeable than a low-salience alert. This could be through a visual alert with higher contrast, an auditory alert that is louder or more descriptive, and/or other standout properties. Higher salience is generally expected to improve driver detection, but this assumption has not been tested across both visual and auditory modalities in a driving context.

The theoretical basis for salience-driven attention capture, as outlined by Theeuwes (2010). Theeuwes reviewed behavioral and neural evidence and argued that the initial visual stage is stimulus-driven and that top-down goals bias selection only later in processing. Salient stimuli capture attention regardless of what the observer is looking for. Under this framework, high-salience visual alerts should quickly capture a driver's attention. What happens after the capture, though, is a separate question.

In a driving environment, the visual scene is already highly complex and constantly moving with every changing variable, which may reduce the direct benefit of high-salience visual alerts. An alert that stands out on a static screen does not necessarily stand out the same way it does on an actual roadway. Einhäuser, Neubert, Grimm, and Bendixen (2024) found that increasing visual salience caused participants to orient their gaze to an alert faster but actually slowed their overall response time. High-salience visual alerts drew attention quickly but then interfered with the execution of the response. Increasing auditory salience did not produce the same problem, rather it improved both the attention to the alert and response speed. The relationship between salience and performance is not straightforward, and the direction of the effect varies by modality.

For auditory salience, loudness is one factor, but Baldwin and May (2011) and Wu et al. (2018) show it is not the only factor. Tonal quality, spectral content, duty cycle, and meaning all contribute to how much an auditory alert draws one's attention. As Wu et al. (2018) show, the highest perceived urgency does not always produce the best behavioral response. How those dimensions interact with modality to affect detection performance is one of the main questions this study hopes to address.

2.4 Visual vs. Auditory Modality in Driving Tasks

Bella and Silvestri (2017) directly compared auditory speech messages and visual icons in a driving simulator at intersections with 41 drivers. Both alert types cut reaction time compared to no warning, but the auditory was the faster of the two. Drivers in the auditory condition reacted in an average of 1.26 seconds, while those in the visual condition took an average time of 1.49 seconds, a difference of 0.23 seconds ($p = 0.018$). Zero collisions occurred in both warning conditions; however, in the no-warning condition, 16 percent of interactions ended in a collision.

Bella and Silvestri (2017) explain the result as visual warnings required drivers to shift their gaze to a display, read the directional icon, and then look back at the road. Whereas auditory warnings allowed drivers to keep their eyes on the road throughout and kept them more aware. The visual alert added a step that the auditory alert did not, and that extra step came at the cost of drivers' reaction time.

Wang, Wu, Ma, Wang, Gan, Liu, and Yang (2022) examined how the mapping. Characteristics of visual and auditory warning constitute an effect on driver detection and performance in a similar driving task with a sample of 32 participants. They found that multimodal warnings outperformed unimodal warnings across conditions, and that high mapping auditory, meaning sounds that closely resemble the real-world sound they represent, cut braking reaction time compared to low mapping auditory warnings. However, high mapping visuals did not produce the same benefit. Eye-movement data showed that complex visual warning graphics diverted drivers' fixation from the road, whereas auditory alerts captured their attention without requiring a visual shift.

Wang et al. (2022) claimed that auditory signals excel at capturing driver attention and increasing perceived urgency, while visual signals are better suited to presenting detailed information. When the primary task already demands continuous visual attention, adding an auditory channel draws on a separate resource pool and avoids the competition that a visual channel creates, thereby reducing the driver’s cognitive load.

Scott and Gray (2008) compared tactile, visual, and auditory rear-end collision warnings in a driving simulator with 16 participants at two warning timings: 3 seconds and 5 seconds before the collision. Modality effects were significant only in the early (5 seconds to crash) warning condition. Tactile warnings produced significantly faster brake reaction time than visual warnings. Auditory warnings fell between but were not significantly different from visual. The modality

advantage disappeared at the late timing (3 seconds to crash). The modality effect depended greatly on the time before the warning.

Lylykangas, Surakka, Salminen, Farooq, and Raisamo (2016) tested 20 participants on visual, tactile, and visual-tactile forward collision warnings while manipulating gaze across four locations (forward roadway, left mirror, center mirror, and right mirror). Brake reaction times to visual warnings were the slowest when drivers were looking at the right mirror, the location furthest from the warning display. Brake reaction times to tactile and visual-tactile warnings were unaffected by gaze location. Visual alert effectiveness depends on where the driver is looking and how long it takes for them to refocus on the warning display.

Gaspar, Schwarz, Brown, and Kang (2019) found a similar effect among drowsy drivers (N=31). Forward collision warnings speed response time only when drivers' eyes were down at the onset of the road hazard. When their eyes were already on the road, the warnings produced no measurable benefit. Both Lylykangas et al. (2016) and Gaspar et al. (2019) show that nonvisual alerts do not pay the gaze position penalty that visual alerts do. This is one reason H3 predicts that there will be an auditory advantage.

Schwarz and Fastenmeier (2017) found that modality effects also interact with design factors. In a between-subjects study with 88 drivers, they crossed alert modality (visual and auditory) with specificity (low vs high information content) for forward collision warnings. The modality's main effect was qualified by a modality-by-specificity interaction; specific visual warnings outperformed specific auditory warnings, while less specific warnings showed the opposite pattern. Modality effects depend on the design parameters with which they interact with. This is exactly why this study will cross-modality with salience rather than test modality solely.

Strayer et al. (2015) developed a Detection Response Task using a 20/80 visual oddball with keyboard responses to measure cognitive distraction across three studies: lab, driving simulations,

and on-road. The DRT has since been formalized as an International Organization for Standardization (ISO) standard for measuring driver cognitive load. The PsychoPy-based oddball detection in this study uses the same kind of design.

These studies support H3's prediction that high salience auditory alerts will outperform high salience visual alerts on both reaction time and accuracy. They also show that the modality advantage is not universal. It depends on warning timing, gaze position, design specificity, and the visual demands of the driving task.

2.5 Validity of Video-Based Driving

This study used pre-recorded driving videos rather than a high-fidelity driving simulator.

Horswill, Hill, and Wetton (2015) conducted a prospective cohort study with 33,105 newly licensed Australian drivers; of those, 5862 were followed for 1 year after taking a video-based hazard perception test. Drivers who failed the video-based test were 25 percent more likely to be involved in an active crash in the following year, controlling for age, gender, and driving exposure. Failing drivers were also 17 percent more likely to have been in active crashes prior to the test. A short video-based driving task predicted real crash risk both before and after the test was completed.

Muela, Chica, Garcia-Fernandez, and Castro (2021) tested whether classic attention findings replicate in realistic driving videos. With 92 participants across three experience levels, they examined the validity of spatial cues (valid, invalid, or no-cue) using a "what happens next" prediction task built from real driving footage. The classic spatial cueing effect also appeared in driving videos. Participants gave better predictions with valid cues than with invalid cues at all three experience levels. There was also no interaction between cue validity and driving experience.

The attentional capture effect was the same for novice, experienced, and non-driver participants. External attention seems to work the same way across all three experience levels when realistic driving videos are used.

Wolfe, Kosovicheva, Stent, and Rosenholtz (2021) tested the same kind of effects with reaction time. Their study comprised 100 participants who viewed prerecorded dashcam driving videos with valid spatiotemporal cues, invalid spatiotemporal cues, temporal-only cues, and a no-cue baseline. Valid spatiotemporal cues speed hazard localization reaction time by approximately 60ms, relative to baseline. Invalid spatiotemporal cues reduced localization accuracy by 11 percent. The reaction time effects were roughly twice as large as the effects from Posner's spatial cueing experience with simple stimuli. Attention cueing effects hold up in dashcam driving videos and may be even larger in a lab task.

Chen, Šabić, Mishler, Parker, and Yamaguchi (2020) is the closest study to this one in terms of design. They ran two experiments (N=42 and N=47) where participants viewed video-based semi-autonomous driving scenes and responded to lateral auditory collision warnings using a steering wheel. Pedestrian collision detection warnings produced faster reaction times than avoidance direction warnings in experiment 1, and this effect was driven by attention capture rather than by stimulus-response compatibility. Video-based driving stimuli, auditory alerts that vary in directional content, and speeded directional responses are all a part of this study’s design.

The video-based models used here are not a substitute for a full vehicle simulation, and the limitations of removing motor control from the participants will be acknowledged in the discussion of this report. These four studies show that they can predict real-world crash risk (Horswill et al. 2015), replicate the same attentional mechanism found in classic lab studies (Muela et al., 2021; Wolfe et al., 2021), and they all serve as valid setups for studying alert detection performance (Chen et al., 2020).

3. Method

3.1 Participants

Twenty-four adult participants (13 women, 11 men) were recruited through in-person invitation, text messages, and outreach on the San Jose State University campus, including classmates, coworkers, and volunteers from the researchers academic and professional networks. Participant ages ranged from 22 to 44 years old (M = 29.3, SD = 5.3). All participants met the eligibility criteria specified in Table 1. Twelve participants were assigned to counterbalance group A, and twelve were assigned to counterbalance group B, with group placement determined by the participant ID number, with group A being odd numbers and B being even. All 24 participants completed the full 34-40 minutes session, and no participant was excluded from analysis. The study protocol was approved the San Jose State University Institutional Review Board (IRB), and all participants were provided their informed consent prior to participation.

Table 1. Participant eligibility criteria

Criteria	Requirement
Age	18 years of age or older
Driving Status	Valid driver's license
Vision	Normal or corrected-to-normal vision
Hearing	Normal hearing
Consent	Able to provide informed consent in English

3.2 Experimental Design

This study uses a 2 × 2 within-subject design in which each participant completes all four alert conditions. The independent variables are alert modality (visual vs auditory) and alert salience

(low vs high). Crossing these produces four conditions, shown in Table 2. A within-subject design was chosen so that participants' responses could be compared directly across conditions without individual differences in reaction time or attention skewing the results.

To control for salience-order effects, the four conditions were counterbalanced across two different run orders. Participants assigned to counterbalance group A completed the conditions in the order of auditory high, visual high, visual low, and auditory low. Participants assigned to counterbalance group B completed the condition in the reverse salience order, auditory-low, visual-low, visual-high, auditory-high. The two driving videos (Chicago and Los Angeles) alternated across the four runs for both groups, so that no single video was paired with a single modality across the sample. Group assignment was determined by the participant ID number, with even-numbered participants assigned to group A and odd-numbered participants assigned to group B. This then produced 12 participants in each counterbalance group.

Table 2: The four alert conditions of this study.

Condition 1: Low salience visual alert	Condition 3: Low salience auditory alert
Condition 2: high salience visual alert	Condition 4: High salience auditory alert

The dependent variables are reaction time, defined as the time between the onset of the alert and the participant's initial keyboard response, and response accuracy, defined as whether the participant correctly responded to the given alert.

3.3 Procedure

Before the session begins, participants review the informed consent and eligibility screening forms. Those who qualify proceed with the experiment in the San José State Human Factors and Ergonomics Laboratory.

The task runs in PsychoPy. Participants sit in front of a computer monitor and watch pre-recorded driving videos, with alert notifications appearing at pre-set intervals by the researcher. Participants press a designated keyboard button as quickly and accurately as possible in accordance with the alert presented. Visual alerts appear on screen, and auditory alerts will play through the computer speakers. Each participant will complete all four conditions.

After the task, participants fill out a brief post-task questionnaire covering noticeability, ease of recognition, modality preference, and perceived distraction. Participants receive a debrief statement at the end of the session. Total session time is approximately 35-40 minutes.

3.4 Data Collection Instruments

Four instruments are used to collect data. The PsychoPy-based experimental task will be the primary tool, automatically recording reaction time and response accuracy for each alert. All data is linked to an anonymous participant ID number. No personally identifying information is to be collected from the participants.

The second tool is the post-task questionnaire, which will include five Likert-scale questions and one open-ended question on noticeability, ease of recognition, modality preference, and perceived distraction. The eligibility screening form confirms participants meet all inclusion criteria before the session begins, and the informed consent form covers the procedures, risks, and the participants' right to stop at any time without penalty.

3.5 Statistical Analysis

Detection performance will be analyzed using a 2 × 2 repeated-measures ANOVA, with separate analyses for reaction time and response accuracy. Alert modality (visual and auditory) and alert salience (high and low) are the within-subject factors. The analysis tests the main effects

of modality and salience, as well as their interaction. All analyses will be run in SPSS at an alpha level of .05.

4. Results

4.1 Data cleaning

A total of 1,008 trials were recorded among the 24 participants (42 trials per participant). Sixty-five trials (6.4%) were classified as missed trials, defined as those in which the participant did not respond within the given response window. These trials were retained in the dataset and analyzed separately as the miss rate, with reaction time set to missing for the reaction-time analysis. No trials were excluded due to audio playback failure, and no outlier trimming was done to the reaction time data. All trials with valid responses were included in the reaction time analysis.

4.2 Reaction Time

Mean reaction times for each of the four conditions are presented below in Table 3. Across all conditions, mean reaction time ranged from 591ms (Visual, high salience) to 757ms (Visual, low salience). Standard deviations were comparable across all conditions, ranging from 68ms to 80ms.

Table 3. Mean reaction time (ms) by modality and salience (N = 24)

Condition	M(ms)	SD (ms)	95% CI lower	95% CI upper
Visual-High	591	80	557	925
Visual Low	757	68	728	786
Auditory-High	619	72	588	649
Auditory- Low	736	75	704	768

A 2 (modality: visual and auditory) x2 (salience: low and high) repeated-measures ANOVA was conducted on the mean reaction time. The main effect of salience was significant, $F(1,23) = 8.69.47$, $p<.001$, partial $\eta^2 = 0.489$. High salience alerts produced faster reaction times than low

salience alerts across both modalities tested. The main effect of modality was not significant, $F(1,23) = 0.72$, $p = .405$, partial $\eta^2 = .001$. The modality × salience interaction, however, was significant, $F(1,23) = 39.55$, $p<.001$, partial $\eta^2 = 0.027$. The full ANOVA results are posted in Table 4 below.

Table 4. Repeated measures ANOVA results for reaction time

Effect	***F***	***df***	***P***	***Partial η²***
Modality	.72	1,23	.405	.001
Salience	869.47	1,23	<.001	.489
Modality × Salience	39.55	1,23	<.001	.027

As shown in Figure 1 below, the means did produce a pattern. The Visual alerts produced the fastest mean reaction time of any condition, while at high salience (591ms), and the slowest reaction time at low salience (757ms). Auditory alerts, however, showed a smaller spread between salience levels, with high-salience alerts averaging 619ms and low-salience alerts averaging 736ms. This interaction reflects the difference in spread; the salience effect was larger for visual, with a spread of 166ms, than for auditory, with a spread of 117ms.

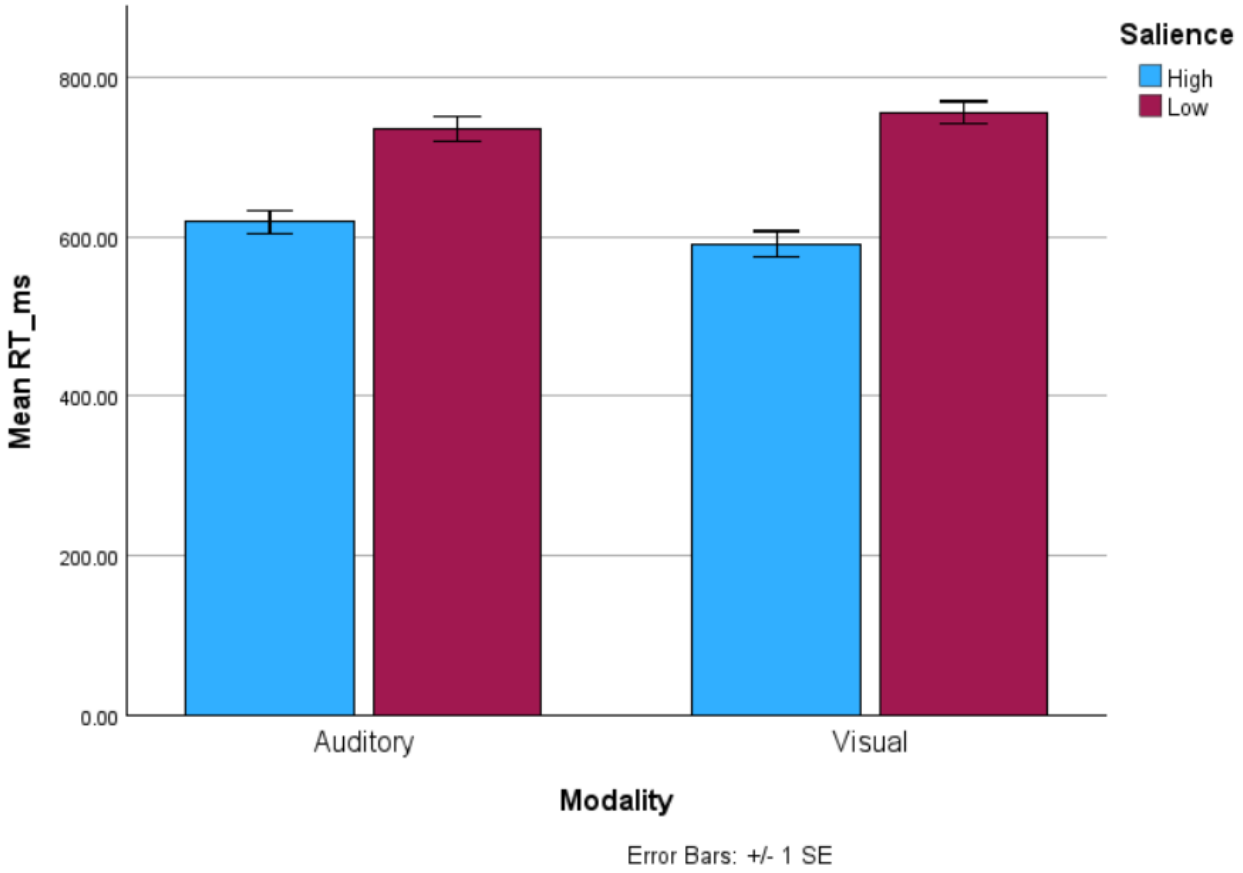


Figure 1. Mean reaction time by modality and salience. Error bars represent ± SEM.

To address the three hypotheses, three planned paired-samples t-tests were conducted with a Bonferroni-corrected alpha of .0167 (.05/3). For H1, the reaction time in the visual-high salience condition was significantly faster than the visual-low salience condition, $t(23) = -28.66$, $p<.001$, $d = -5.85$. For H2, the reaction time in the auditory-high salience condition was also significantly faster than in the auditory-low salience condition, $t(23) = -18$, $p<.001$, $d = -3.67$. Finally, for H3, reaction time in the visual-high salience condition was also significantly faster than the auditory-high salience condition, $t(23)$, $p<.001$, $d = -1.05$. The reaction time data supported all three hypotheses, and the results of the paired comparison are summarized in Table 5 below.

Table 5. Planned paired-samples t-tests for the three hypotheses (Bonferroni alpha =.0167).

Hypothesis	Comparison	*t*(23)	*p*	Cohen's *d*
H1	Visual-high vs Visual-low	-28.66	< .001	-5.85
H2	Auditory-High vs Auditory-Low	-18	< .001	-3.67
H3	Visual-high vs Auditory-high	-5.12	< .001	-1.05

Shapiro-Wilk tests of normality were conducted on the reaction-time data for each of the conditions tested. Three of the four conditions were consistent with normality (visual-high, W = .97, p=.569; visual-low, W = .92, p= .066; auditory-high, W = .98, p = .845). The auditory-low condition deviated from normality (W = .87, p = .004).

4.3 Miss Rate

Miss rate was calculated as the proportion of trials per condition in which the participant did not respond within the response window. Mean miss rates ranged from 3.1% (auditory-high) to 9.3% (visual-low). Descriptive statistics for miss rates are in Table 6 below.

Table 6. Mean miss rate (proportion misses) by modality and salience condition (N=24)

Condition	*M*	*SD*	95% CI Lower	95% CI Upper
Visual-high	.038	.062	.012	.064
Visual-low	.093	.082	.058	.127
Auditory-high	.031	.056	.007	.055
Auditory-low	.090	.080	.057	.124

A 2 (modality) x 2(salience) repeated measures ANOVA was conducted on miss rate. The main effect of salience was found to be significant, $F(1,23) = 14.39$, $p<.001$, partial $\eta^2 = .143$.

Low- salience alerts were missed more often than high-salience alerts in both modalities tested. The main effect of modality was not significant, F (1,23) = 0.11, p = .743, partial η^2 = .001. The modality × salience interaction was also not significant, F (1,23) = .002, p = .889, partial η^2 =000. The full ANOVA results are presented in Table 7 below. The pattern of mean miss rate is shown in Figure 2.

Table 7. Repeated measures ANOVA results for miss rate

Effect	***F***	***df***	***p***	**Partial η^2**
Modality	.011	1,23	.743	.001
Salience	14.39	1,23	<.001	.143
Modality × salience	.002	1,23	.889	0

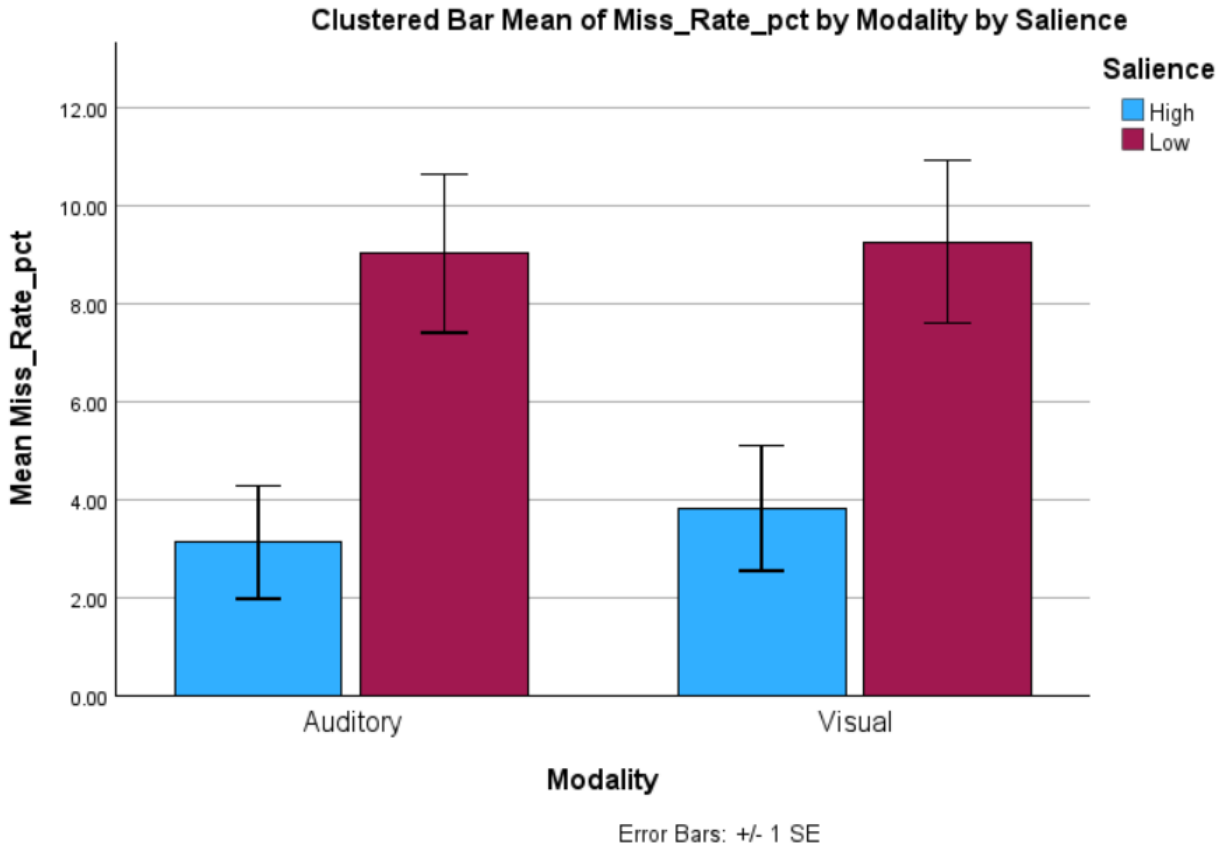


Figure 2. Mean miss rate (%) by modality and salience. Error bars represent ± SEM.

Shapiro-Wilk tests indicated that the miss rate distributions deviated from normality in all four conditions tested. This was expected given that the muss rate is a bounded proportion with a low base rate, which produced a positively skewed distribution in the data. As a sensitivity check, the pattern of effects in the ANOVA mirrors the descriptive pattern in Table 6, with low-salience alerts having roughly twice the miss rate of their high-salience counterparts.

4.4 Response accuracy

Response accuracy was calculated as the proportion of correct keystrokes out of trials in which the participants responded within the given response window. Trials classified as misses were not included in the accuracy calculation, since accuracy and miss rate are different, distinct outcomes (correct identification vs detection). Mean response accuracies for each of the four conditions are

presented below in Table 8. Across all conditions, mean accuracy ranged from 95.9% (auditory-high salience) to 98.4% (visual-high salience).

Table 8. Mean response accuracy (proportion correct of responded trials) by modality and salience condition (N=24)

Condition	*M*	*SD*	95% CI Lower	95% CI Upper
Visual-high	.984	.036	.969	1.0
Visual-low	.981	.044	.962	.99
Auditory-high	.959	.060	.934	.984
Auditory-low	.963	.073	.933	.994

A 2 (modality) x (salience) repeated measures ANOVA was conducted on response accuracy. None of the tested effects were statistically significant. The main effect of modality, although it almost but ultimately did not reach significance, $F(1,23) = 3.35$, $p = .80$, partial $\eta^2 = .037$, with visual alerts producing slightly higher accuracy than auditory alerts. The main effect of salience was not significant, $F(1,23) = .001$, $p = .974$, partial $\eta^2 = 0$. The modality × salience interaction was also not significant, $F(1,23) = 0.13$, $p = .720$, partial $\eta^2 = .001$. The full ANOVA results are in Table 9 below.

Table 9. Repeated measures ANOVA results for response accuracy.

Effect	*F*	*df*	*p*	partial η^2
Modality	3.35	1,23	.08	.037
Salience	.001	1,23	.974	0
Modality × Salience	.13	1,23	.720	.001

Response accuracy was at or near the ceiling for all four conditions tested, with an overall mean accuracy of 97.2% across the 943 responded trials. Of the 24 participants tested, 20 had 100% accuracy in the visual-high condition, 20 in the visual-low condition, 18 in the auditory-low condition, and 15 in the auditory-high condition. Shapiro-Wilk tests confirmed that the accuracy distributions deviated from normality in all four conditions tested (all $p < .001$), which is to be expected when scores are near the ceiling. Given this ceiling effect, the absence of statistically significant effects on response accuracy is to be interpreted with caution. The ANOVA results are reported as the primary analysis, with the understanding that the test had limited sensitivity to detect differences when most participants performed perfectly. The implication of this ceiling effect for the hypotheses is addressed in the discussion of this report.

4.5 Post Task Questionnaire

All 24 participants completed the five Likert-scale questions in the post-task questionnaire; 18 provided additional written responses in the open-ended section, and 6 left it blank. Descriptive statistics for the five Likert rating questions are presented below in Table 10, and the frequency distributions for each item are shown in Figures 5 through 9.

Table 10. Descriptive statistics for post-task questionnaire ratings (N=24)

Question	*M*	*Mdn*	*SD*	Min	Max
Q1: Alert noticeability (1-5	4.04	4.0	1.16	1	5
Q2: Recognition ease (1-5	3.75	4.0	0.90	2	5
Q3: Most noticeable alert	-	-	-	-	-

(categorical)					
Q4: Alert preference(categorical)	-	-	-	-	-
Q5: Alert Distraction (1-5)	1.96	2.0	0.81	1	4

Note. Q3 and Q4 used categorical responses response options without ordinal scaling, so means and standard deviations are not reported for those items. Frequency distributions for these items are reported in the text.

4.5.1 Noticeability and Recognition

Twenty-one of the twenty-four participants (87.5%) rated the alerts as either very or extremely noticeable in Q1, with 12 (50%) rating them very. Two participants (8.3%) rated the alerts as not noticeable, and one (4.2) rated them as slightly noticeable.

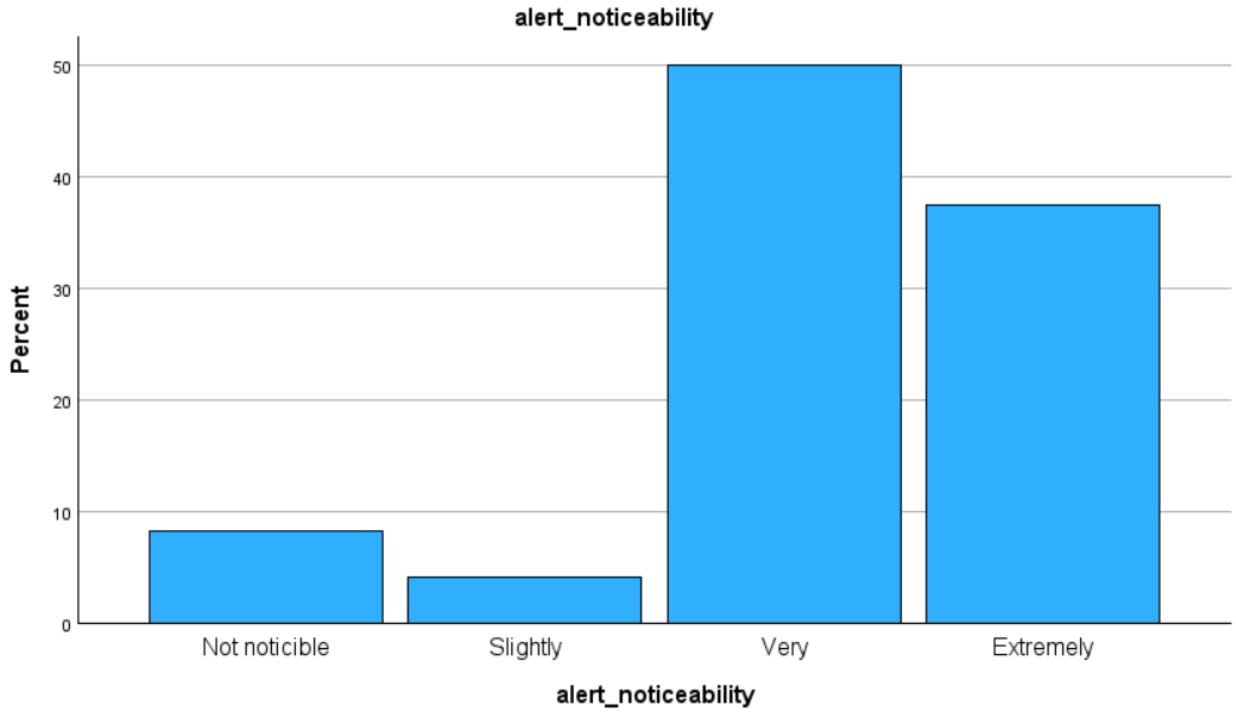


Figure 3. Frequency distribution of responses to Q1 (N=24).

For Q2, fifteen participants (65.2%) rated alert recognition as easy or very easy, with easy as the most popular at 10 responses (41.7%). Seven participants (29.2%) responded neutrally, and two (8.3%) responded that it was difficult. Out of the twenty-four participants, none selected very difficult.

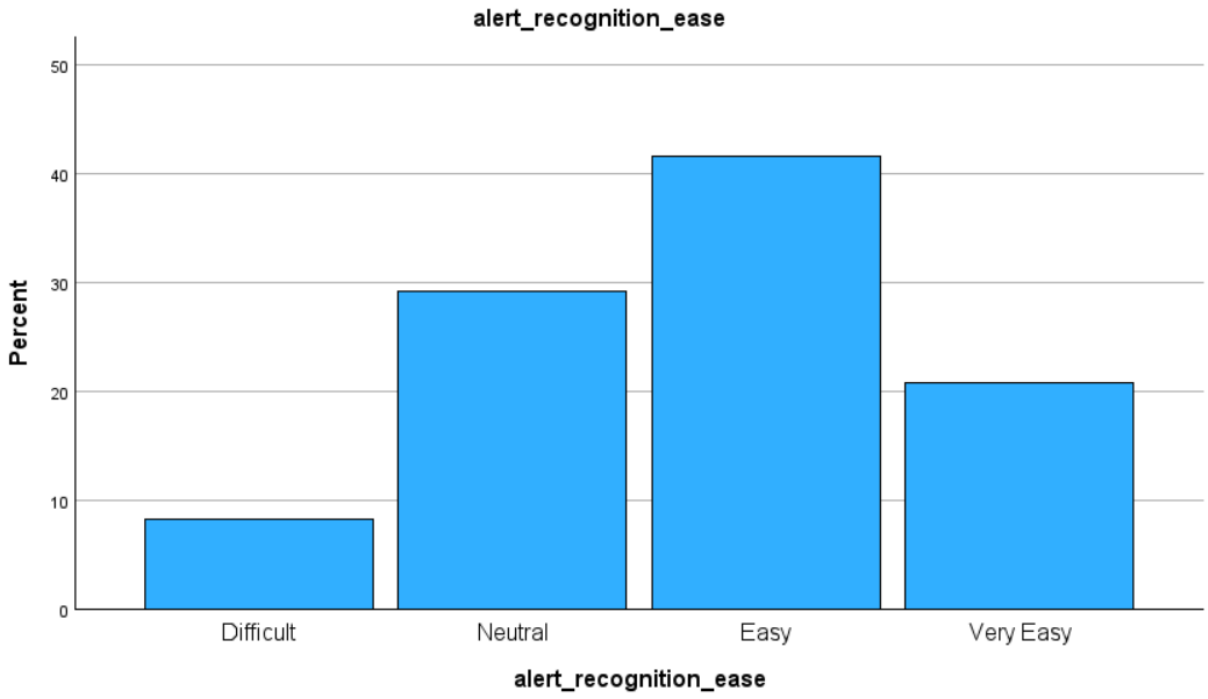


Figure 4. Frequency distribution of responses to Q2 (N=24).

4.5.2 Modality Identification and Preference

Responses to Q3 (which alert type was most noticeable) were evenly split between visual and auditory. Eight participants (33.3%) selected visual, eight (33.3%) selected visual, six (25%) selected both equally, and only two (8.3%) selected not sure.

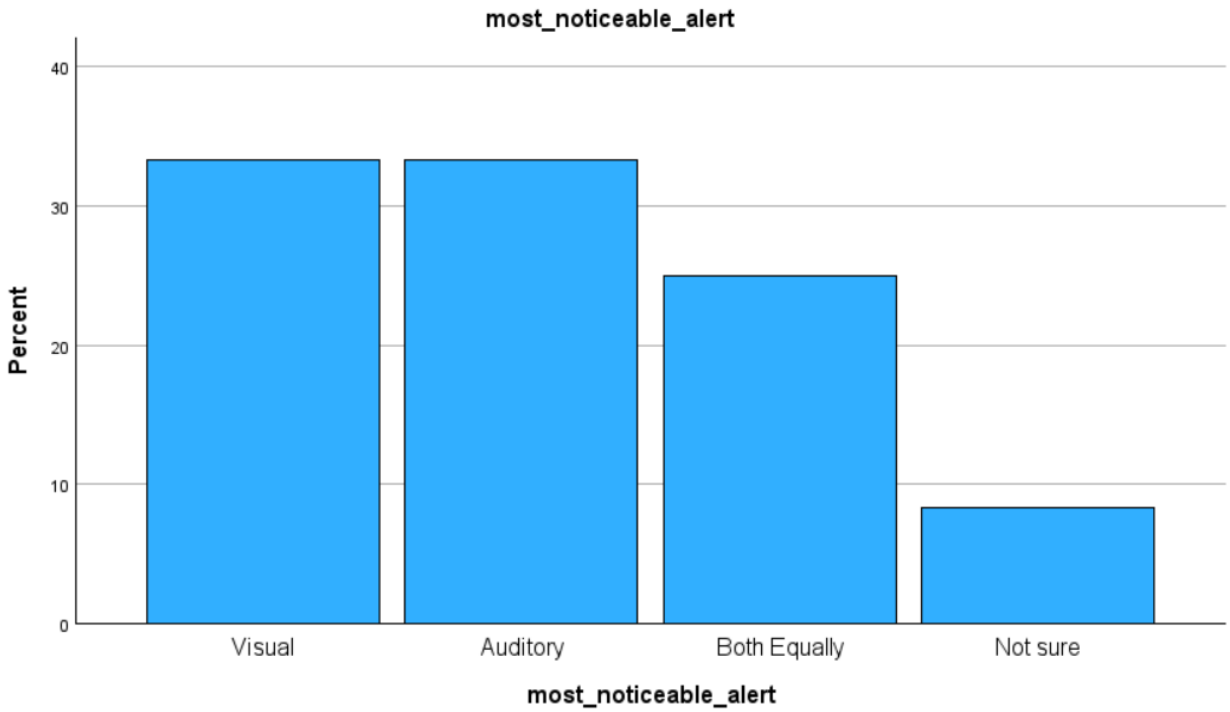


Figure 5. Frequency of distribution response to Q3 (N=24).

The pattern was similar to Q4 (which alert type was preferred). Seven Participants (29.2%) selected visual, seven (29.2%) auditory, six (25%) selected both equally, and four (16.7%) selected no preference.

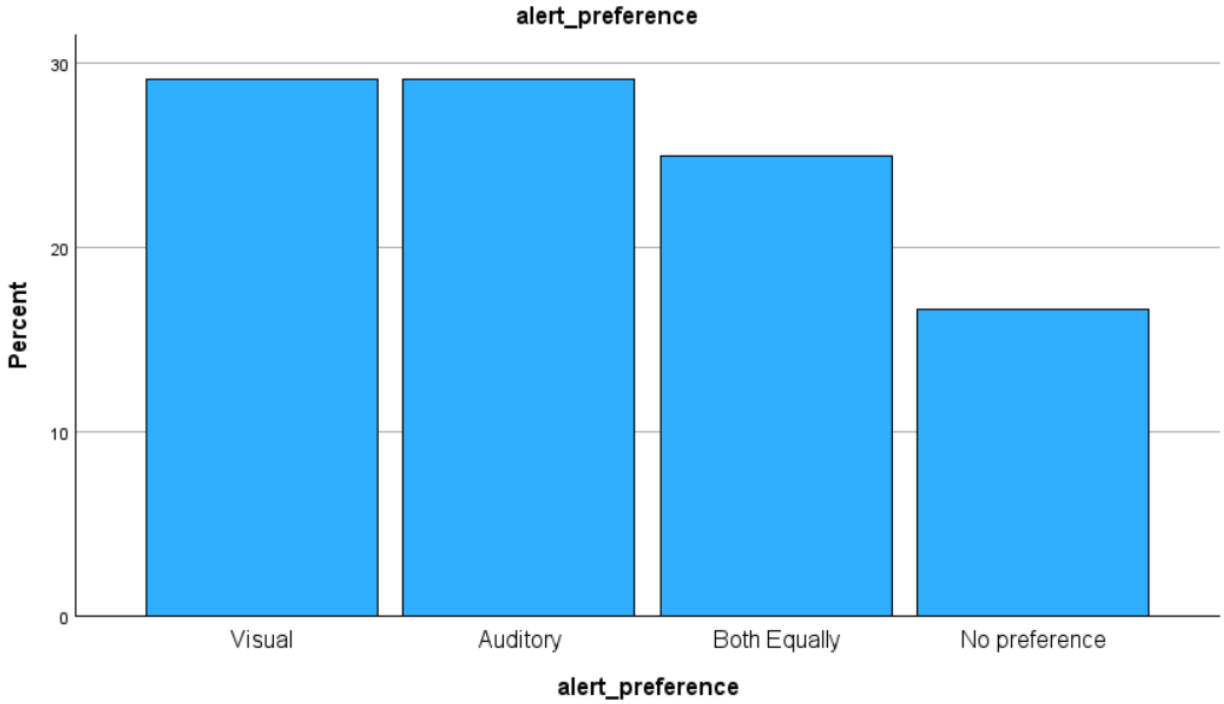


Figure 6. Frequency distribution of responses to Q4. (N=24)

4.5.3 Perceived Distraction

On Q5, twelve participants (50%) rated the alerts as slightly distracting, and seven (29.2%) rated them as not distracting at all. Four participants (16.7%) selected moderately distracting, and one (4.2%) selected very distracting. The mean rating of 1.96 (SD = .81) and the median of 2 indicate that perceived distraction was generally low across the participants.

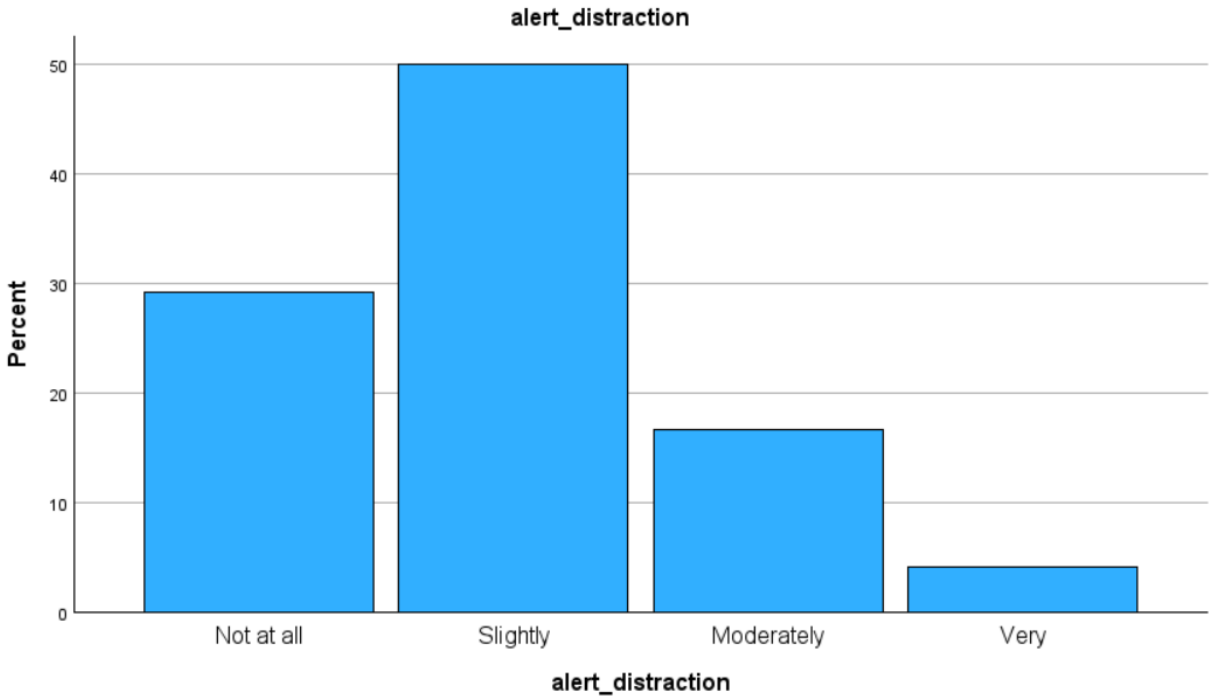


Figure 7. Frequency distribution of responses to Q5 (N=24)

4.5.4 Open-Ended Responses

Eighteen participants (75%) provided written feedback on the open-ended responses section. The full set of responses is presented in Appendix B. Recurring themes across the responses included a preference for the more descriptive, longer auditory phrasing over the shorter, low salience phrasing. Comments also noted that the high-salience visual alerts were easier to detect than the low-salience visual alerts, and that some of the text-message-style alerts felt different from the navigation-style alerts. Several participants noted that the session length felt appropriate and did not introduce fatigue.

4.6 Summary

The salience manipulation produced large and consistent effects on detection performance. Reaction time was significantly faster, and miss rate was significantly lower for high-salience alerts than for low-salience alerts in both modalities tested. The main effect of modality was not statistically significant for any of the three dependent variables, though a non-significant trend favoring visual alerts was observed with the visual alerts for response accuracy. The modality × salience interaction was significant for reaction time, but not for miss rate or response accuracy, with the salience effect being larger for visual alerts than for auditory alerts on the reaction time measure. Response accuracy was at or near the ceiling across all four conditions and did not differ significantly by modality, salience, or their interaction. All three of the hypotheses were supported by the reaction-time analysis. The hypotheses were partially supported by the miss-rate analysis, which showed the predicted salience effect, but no significant modality difference. Questionnaire responses showed that the alerts were generally noticeable, easy to recognize, and only slightly distracting, with participants roughly evenly split between visual and auditory in their selected preference.

5. Discussion

5.1 Summary of Primary Findings

The salience manipulation produced a large and consistent effect on detection performance. High-salience alerts produced faster reaction times and lower miss rates than lower salience alerts in both visual and auditory modalities. Reaction time differed by an average of 166ms between high and low salience visual alerts, and by an average of 117ms between high and low salience auditory alerts. Miss rates were roughly twice as high for low-salience alerts as for high-salience alerts in both modalities tested. The main effect of modality was not statistically significant for any of the three dependent variables. The modality × salience interaction, however, was significant for reaction time, with the salience effect larger for visual alerts than for auditory alerts. Response accuracy was at or near the ceiling across all four conditions and did not differ significantly by either modality, salience, or their interaction.

5.2 Hypothesis 1: Visual High vs Visual Low Salience

H1 predicted that faster reaction times and higher accuracy ratings would be obtained for visual-high salience alerts than for visual-low salience alerts. The reaction time data supported a prediction with a large effect size ($d = -5.85$), with a mean reaction time of roughly 166ms faster in the high salience condition. This was further supported by the miss rate data, with visual-low salience alerts missed nearly twice as often as the visual-high salience counterparts. Response accuracy was at or near the ceiling in both conditions and did not differ significantly between them, so accuracy alone could not distinguish between the two salience levels.

The reaction time and miss rate results align with the predictions derived from Cognitive Load Theory (Paas & Van Merrienboer, 2020). A more salient visual alert demands fewer perceptual

resources to detect against the driving environment, leaving more capacity available for the response decision. Lower-salience alerts blend into the visually complex environment and require additional processing to be identified, increasing both the time to respond and the likelihood that the alert is missed entirely.

5.3 Hypothesis 2: Auditory High vs Auditory Low Salience

H2 predicted faster reaction times and higher accuracy for high-salience auditory alerts than for low-salience auditory alerts. The reaction time data supported this prediction with a large effect ($d = -3.67$), and the miss rate data continued to support this as well, with low-salience auditory alerts missed roughly three times as much as their high-salience counterparts. Accuracy did not differ significantly between the two auditory salience levels, and this was again due to the ceiling effect taking place.

The salience manipulation in the auditory trials was based on phrase content and complexity rather than volume, with high-salience alerts using longer, more descriptive language than the brief commands in the low-salience condition. ‘The reaction time advantage for the longer alerts is consistent with Nees et al. (2016), who found that speech-based auditory alerts that carry verbal information produced better processing of alerted events than brief tones and messages. The same can be seen in Baldwin and May (2011), who showed that auditory warning effectiveness depends on more than volume alone, with semantic content and other auditory parameters contributing to the strength with which an alert is processed. The results in this study are consistent and align with both of their findings.

5.4 Hypothesis 3: Visual High vs Auditory High Salience

H3 predicted faster reaction times and higher accuracy for visual-high salience alerts than for auditory-high salience alerts. This prediction was supported by the reaction time data (d = -1.05), with visual-high salience producing the fastest mean reaction time across the four conditions at 591ms. Miss rates did not differ significantly between the two high-salience conditions, and accuracy was at the ceiling for both.

This finding is the most surprising of the three from a literature perspective. Bella and Silvestri (2017) reported that auditory alerts produced faster reaction times when compared to visual alerts in a simulated intersection task, with auditory alerts averaging 1.26 seconds and visual alerts averaging 1.49 seconds. The rationale offered was that visual alerts required drivers to shift their gaze away from the road to interpret the icon, adding processing steps that the auditory system did not have. The results of the current study show the opposite pattern at high salience.

A likely explanation is the physical location of the visual alerts in this study. Bella and Silvestri presented visual icons on a separate display that required participants to shift their gaze from the road, which is how visual alerts were presented in this study. In this study, the visual alerts were presented as an overlay on the same screen as the driving video, in the approximate foveal region of participants' visual attention. This placement removes the gaze-shift cost in time that disadvantaged visual alerts in Bella and Silvestri's study, aligning the visual alert closer to where the driver was already looking. Given this placement, the high-salience visual alerts could be detected within the existing visual fixation, while the auditory alert still required attention from the visual scene to the auditory channel before the response could be initiated and executed.

This interpretation is consistent with Ho et al. (2007), who showed that the relative advantage of one modality over another depends heavily on the spatial and temporal relationship between the

alert and primary assigned task. When a visual alert is presented near the existing point of attention, the cost of using it is reduced.

5.5 The Modality × Salience interaction

The interaction between modality and salience on reaction time was statistically significant. The size of the salience effect was larger for visual alerts (166ms) than for auditory alerts (117ms). At high salience, visual alerts produced the fastest reaction time of any condition, but at low salience, visual alerts produced the slowest reaction time of any condition.

This pattern is consistent with the findings seen in Einhäuser et al. (2024), who reported that the relationship between visual salience and detection performance is non-monotonic and modality specific. In their study, increasing visual salience drew gaze to the alert quickly, but disrupted the response. Increasing auditory salience improved both the attention and the response speed. The current study builds on this, when visual salience is sufficient to support detection within the foveal region, visual alerts outperform auditory alerts, and when visual salience is insufficient, the visual alerts become the slowest of the conditions because the alert is then competing with a visually rich and complex driving environment that is already consuming the participant's perceptual resources. The Auditory alerts do not face this competition (Wang et al., 2022), so the salience effect in the auditory modality is smaller.

Visual alerts are more sensitive to salience manipulations than auditory alerts.

This pattern has clear implications for IVIS design; a poorly tuned visual alert can perform worse than a poorly tuned auditory alert, even though a well-tuned visual alert can outperform any auditory alert.

5.6 The Ceiling Effect on Response Accuracy

The hypotheses for this study were framed in terms of both reaction time and response accuracy. The reaction time analysis supported all three hypotheses with large effect sizes; however, the response accuracy did not produce statistically significant effects for any of the three predictions. The most likely explanation is a ceiling effect. Response accuracy averaged at 97.2% across all responded trials, with 20 of the 24 participants achieving 100% accuracy in the visual-high condition, and 18 out of 24 achieving 100% accuracy in the auditory-low condition. When the bulk of the sample is performing at the upper bound of the scale, the test then has limited statistical sensitivity to detect differences between the conditions, regardless of whether such differences exist within the study population. The directional hypotheses for accuracy were therefore not refutable by the data, as they were untestable with the sample and task design used for this study.

A more realistic interpretation is that the four key response mappings used in the study did not use identification or motor execution at the level of difficulty produced by any of the four alert conditions. Once an alert was detected, participants identified the correct response with near-perfect accuracy across all conditions. This is, however, an informative result, as the salience manipulation affected the detection stage of processing (as reflected in reaction time and miss rate) without affecting the identification stage (as reflected in the accuracy scores). The implication is that, in this task, perceptual salience determined whether an alert was caught in time, not whether the participant knew what to do once they had become aware of it.

The accuracy-dependent variable did not contribute meaningful information beyond what the reaction time and miss rate analyses already provided in the sections prior. Future studies that wish

to test accuracy effects should use a more demanding response mapping or a more time-constrained response window to move performance away from ceiling effects.

5.7 Subjective Preferences and Objective Performances

The questionnaire results showed a notable disconnect between participants stated preferences and their measured performance. On Q3 (which alert type was most noticeable), responses were almost evenly split between visual (33.3%) and auditory (33.3%). On Q4 (which alert type was preferred), 29.2% chose visual, 29.2% chose auditory, and the remaining 41.% selected no preference or both. When asked directly, the participants did not perceive a clear difference in modality in the alerts they had just experienced.

However, the objective data shows a very different pattern. High-salience visual alerts produced the fastest reaction times of any condition tested, and the salience effect on reaction time was larger for the visual channel than for the auditory channel. This dissociation has been reported elsewhere in alert literature. Subjective ratings of alert noticeability and preference do not always align with objective measures of detection performance, particularly when the performance differences are small in absolute terms. For IVIS designers, this is a meaningful point to be aware of; relying on user preference data alone to choose between alert modalities is likely to miss the real differences in detection performance that emerge in behavioral data. Both objective and subjective measures should be taken into account for future design decisions.

5.8 Practical Implications for IVIS Design

The findings of this study show three practical implications. First, the salience of an IVIS alert is the most important factor in determining how quickly drivers respond. Designers should prioritize getting alert salience right over choosing between auditory or visual delivery. The

salience effect was significantly larger than the modality effect in terms of reaction time and miss rate.

Second, when visual alerts are placed near the driver's existing point of fixation and rendered with sufficient salience, they can outperform auditory alerts. The finding that auditory alerts have an advantage in driving tasks (Bella & Silvestri, 2017; Wang et al., 2022) appears to heavily depend on the spatial placement of the visual alert. Visual alerts that require a gaze shift away from the road are likely to be slower than auditory alerts, but visual alerts integrated into the central visual scene do not exhibit this disadvantage.

Third, low salience visual alerts are the worst performers in this study. A poorly tuned visual alert competes with the visually rich, complex driving environment for perceptual resources and can be slower than any other alert type tested. If a designer is unsure whether a visual alert will be sufficiently salient in a driving environment, an auditory alert may be the safer option, as the auditory channel showed less sensitivity to the salience manipulation.

5.9 Limitations of the study

While interpreting the results, several limitations should be considered. First, the study used a within-subject design with N = 24. This sample size was sufficient to detect large effects observed for reaction time and miss rate, but was likely underpowered to detect smaller effects, including the trend toward higher accuracy for visual alerts ($p=.080$) that did not reach statistical significance. A larger sample would most likely clarify if this trend reflected a real effect.

Second, the task is used by video-based driving rather than a simulator or real driving. While video-based driving has been used in the alert detection literature (Nees et al., 2016) and captures the perceptual demands of driving with reasonable fidelity, it does not capture the motor and motivational demands of having actual vehicle control. Reaction times measured in this study

likely reflect detection performance under reduced workload compared to active driving, and the absolute values should not be directly compared to studies using simulators or real vehicles.

Third, the response keys (left, right, up, and space) used a standard keyboard rather than a steering wheel-mounted control. The motor response in real driving would involve a different set of effectors and would likely produce slower reaction times.

Fourth, response accuracy reached a ceiling across all four tested conditions, limiting this DV's ability to distinguish among conditions. As discussed in section 5.6, the accuracy hypotheses were untestable in this sample rather than refuted.

Fifth, the auditory salience manipulation operated through phrase length and directiveness rather than loudness, whereas the visual salience manipulation operated through brightness, size, and color contrast. The two salience manipulations are therefore not directly comparable. The results showed that both manipulations affected detection performance, but the magnitude of the effects cannot be equated across modalities.

Sixth, the distribution of miss rate scores deviated from normality in all four conditions tested, as expected for a bounded proportion with a low base rate. The ANOVA was sufficient to detect the main effect of salience, but a generalized mixed-model approach may have more accurately captured smaller effects on miss rate.

Finally, the two driving videos used (Chicago and Los Angeles) were not equated for visual or auditory complexity. Counterbalancing the videos across modalities reduced, but did not eliminate, the possibility that scene-specific properties contributed to the observed effects.

5.10 Ideas and Directions for Future Research

Several directions for future research follow from the findings and limitations of this study. The most direct extension is a replication with a larger sample and a response task that does not

produce a ceiling effect on accuracy. A more challenging response mapping or shorter window would move accuracy off the ceiling and allow direct testing of accuracy prediction in all three hypotheses.

A second direction is to vary the spatial placement of the visual alerts systematically. The interpretation in section 5.4 claims that the visual alerts in this study were close to the driver's fixation point, and that this placement eliminated the gaze-shift cost reported by Bella and Silvestri (2017). A study that manipulated visual alert placement (foveal vs peripheral vs dashboard) within a single experiment would provide a direct test of this account

A third direction is to extend the salience manipulation to more levels. The current study used a simple high-low contrast in both modalities tested. Salience manipulation across three or more levels would allow for a more precise estimate of the salience-performance function and would test whether the relationship is linear or shows the non-monotonic shape reported by Einhäuser et al. (2024).

A fourth direction is to test these effects in a driving simulator or in a real vehicle equipped with sensors in a test environment. The video-based task used here is well-suited for early-stage detection performance work but cannot speak to how the salience and modality effects interact with active vehicle control. A driving simulator study with the same alert conditions would clarify whether the patterns observed here generalize to real-world driving demands.

Finally, a fifth direction to examine is where the disconnect between subjective and objective performance observed in this study persists across different IVIS contexts. If users consistently fail to identify the modality that produces better detection performance, this has implications for how IVIS systems are evaluated in user research and consumer testing.

6. Conclusion

6.1 Restatement of the Problem and Objectives

In-vehicle information systems (IVIS) present drivers with a steady stream of visual and auditory alerts that compete with the primary task of driving. The effectiveness of these alerts depends heavily on how quickly drivers detect them, how often they miss them, and how accurately they respond. The literature on alert modality and salience contains conflicting findings regarding which modality yields better detection performance and how alert salience interacts with its modality. The objective of this study was to test how alert modality (visual vs auditory) and alert salience (low vs high) affect detection performance while completing a video-based driving task, using erection time, miss rate, and response accuracy as dependent measures.

The study addresses three hypotheses: High salience visual alerts would outperform low salience visual alerts (H1); high salience auditory alerts would outperform low salience auditory alerts (H2); and high salience visual alerts would outperform high salience auditory alerts (H3).

6.2 Summary of Key Findings

Twenty-four participants completed four runs of a video-based driving task, with one run for each condition of modality and salience.

The salience manipulation produced large effects on detection performance. Reaction time was 166ms faster for high salience visual alerts than for low salience visual alerts, and 117ms faster for high salience auditory alerts than for low salience auditory alerts. Miss rates were roughly twice as high for low salience when compared to high salience in both modalities tested. The main effect of salience was statistically significant for both reaction time, $F(1,23) = 869.47$, $p<.001$, partial $\eta^2 = .48$, and miss rate, $F(1,23) = 14.39$, $p<.001$, partial $\eta^2 = .143$.

The modality manipulation, however, did not produce a statistically significant main effect on any dependent variable. Though a significant modality × salience interaction was observed for reaction time, $F(1,23) = 39.55$, $p<.001$, partial $\eta^2 = .027$. The salience effect was larger for visual alerts (166ms) than for auditory alerts (117ms). Visual alerts produced faster reaction time for any condition at high salience (591ms) and the slowest reaction time of any condition at low salience (757ms)

All three hypotheses were supported by the reaction time data, with large effect sizes for H1 ($d = -5.85$), H2 ($d = -3.67$), and H3 ($d = -1.05$). The hypotheses were also partially supported by the miss rate data, which showed the predicted salience effect but no significant modality difference. Response accuracy reached the ceiling across all four conditions and could not be used to evaluate the hypotheses.

The post-task questionnaire showed that participants generally rated alerts as noticeable, easy to recognize, and only slightly distracting. Participant preference for visual versus auditory alerts was nearly evenly split, despite a clear performance advantage for visual alerts at high salience.

6.3 Conclusions

The goals of this study were achieved, as the study produced a complete dataset across all four conditions tested, addressed each of the three hypotheses with appropriate tests, and identified a clear pattern of results that can be used to inform IVIS design.

Three main conclusions are made from the findings, with the first being that alert salience is the dominant factor in determining how quickly drivers respond to IVIS alerts. The salience effect of reaction time was significantly larger than the modality effect. Designers should treat salience as the highest-priority design parameter, even before deciding on the alert's modality.

Second, the relative advantage of visual versus auditory alerts depends on whether the visual alert is presented near the driver's existing point of fixation/attention. When visual alerts are integrated into the central visual scene and given sufficient salience, they tend to outperform auditory alerts. However, when visual alerts are not sufficiently salient, they then become one of the slowest alert modalities. The auditory channel showed less sensitivity to the salience manipulation and is the safer default choice when the salience of alertness cannot be guaranteed.

Third, subjective preference data did not reflect objective performance differences observed in the same participants. User preference research alone is unlikely to be able to identify the modality that produces better detection performance. Behavioral measures should be used alongside subjective measures when evaluating IVIS alert designs.

6.4 Contributions

This study contributes to IVIS alert design literature in three key ways. First, it provides direct experimental evidence that the salience effect is larger and more reliable than the modality effect, as measured by both reaction time and miss rate. Most of the existing alert literature reports either a modality or a salience effect. However, few studies have directly compared the actual magnitude of the two effects within the same experimental design.

Second, it identifies a modality × salience interaction that has not been clearly discussed in previous IVIS literature. The finding that visual alerts are more sensitive to salience than auditory alerts has practical implications for how alert systems should be tuned across the two modalities.

Third, it documents a dissociation between subjective alert preferences and objective detection performance in a within-subjects design, adding to the small body of evidence that cautions against relying on user preference data alone in an alert system evaluation.

6.5 Recommendations

Based on the conclusions in section 6.4, the following recommendations are made for IVIS alert design and evaluation:

For alert design, designers should prioritize salience over modality when configuring new IVIS alerts. High salience alerts produce faster reaction times and lower miss rates across modalities, and the cost of getting salience wrong is higher in the visual channel than in the auditory channel.

For visual alert placement, designers should render visual alerts close to the driver's existing point of fixation, rather than on a separate display that requires the driver to shift their gaze. Visual alerts placed in a central location, such as a heads-up display (HUD), and given sufficient salience, can outperform a highly salient auditory alert.

Lastly, for evaluation, designers and researchers should pair subjective preference assessments with objective performance measurements when evaluating alert systems. User preferences in this study did not align with the performance differences observed in the same users, and either type of measurement alone would have failed to produce the complete picture.

6.6 Future Work

There are several ways future work would broaden and strengthen the conclusions of this study. The most direct approach is to replicate with a larger sample and a more demanding response task that does not produce a ceiling effect on accuracy, thereby allowing a direct test of the accuracy predictions that could not be evaluated in this study. A systematic manipulation of visual alert placement, comparing foveal, peripheral, and dashboard locations within a single experiment, would test the spatial placement account offered within the findings of H3. Extending the salience manipulation to more than two levels would also enable a finer, more accurate estimate of the

salience-performance relationship and better test whether the relationship is linear or non-monotonic across modalities. Replicating these effects in a driving simulator or real vehicle would also clarify whether the patterns observed here hold under the additional motor and motivation demands of active vehicle control. Finally, expanding the questionnaire component would allow a more detailed examination of why users' subjective experience of alerts does not align with their objective detection performance, and whether this dissociation generalizes across different IVIS contexts.

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Appendix A: Terminology and Abbreviations

A.1 Abbreviations and Definitions

Abreviations	Meaning
ANOVA	Analysis of Variance
CI	Confidence Interval
d	Cohen's d (effect size for paired comparisons
df	Degrees of freedom
F	F-statistic (ANOVA test statistic)
H1, H2, H3	Hypotheses 1, 2, and 3
IVIS	In-Vehicle Information System
M	Mean
Mdn	Median
ms	Miliseconds
N	Total sample size
n	Subgroup sample size
p	Probability value (significance level)
Partial η^2	Partial Eta-squared (effect size for ANOVA)
RT	Reaction time
SD	Standard deviation
SEM	Standard Error of the mean
t	T-statistic (test statistic for t-test
TTS	Text to speech
W	W-statistic (Shapiro-Wilk normality test)
HUD	Heads up display

A.2 Terminology and Concepts

Accuracy: In this study, the proportion of trials in which the participant pressed the correct response key out of trials in which a response was made within the response window. Accuracy does not include missed trials.

Alert: A signal delivered through the IVIS to convey information to the driver, such as navigation instructions or incoming messages. In this study, alerts were delivered in both auditory and visual formats.

ANOVA (Analysis of Variance): A statistical test that compares the means of two or more groups or conditions. Repeated measures ANOVA is used when the same participants experience all conditions, such as in this study.

Auditory Modality: Information delivered through the sense of hearing. In this study, audio alerts were spoken phrases generated by text-to-speech software.

Ceiling Effect: A measurement limitation that occurs when most participants score at or near the maximum possible value on a test, leaving little room for differences between conditions.

Cognitive Load Theory: A theoretical framework proposing that working memory has a limited capacity, and that learning and performance suffer when demands exceed said capacity.

Foveal Region: The center part of the visual field where visual acuity is at its highest.

In-Vehicle Information System (IVIS): Any system in a vehicle that delivers information to the driver, including but not limited to infotainment systems, navigation systems, and driver assistance interfaces.

Miss/Miss Rate: A miss is when the participant did not respond to the IVIS alert within the allotted response window. Miss rate is the proportion of trials per condition that were missed. Miss

rate differs from accuracy, which measures correctness given that the participant provided a response.

PsychoPy: An open-source Python-based software for running behavioral experiments.

Reaction Time: The elapsed time from the onset of an alert to the participant's initial response, measured in seconds (or milliseconds in displayed values)

Response Window: The time interval during which a participant's keystroke is counted as a response to a given alert. Trials in which no response was made were coded as a miss.

Salience: The degree to which an alert stands out from its background.

Visual Modality: Information delivered through the sense of sight. In this study, visual alerts were rendered as overlays on the same screen as the driving video.

Appendix B: Open-ended Responses

Participant ID	Response
1	“The high salience alerts were way easier to notice.”
2	“I found the detailed audio alerts more helpful because they told me exactly what to do.”
3	“Some of the text message alerts felt distracting because they did not actually tell me what to do next”
4	-
5	“I think I responded faster to the visual alerts, but preferred the auditory ones overall, it had a really natural sounding voice.”
6	“Red icons stood out right away. The dimmer ones blended in with the video to much
7	-
8	“Session length felt about right, nothing felt overwhelming.”
9	“Some alerts came close together and it was hard to process both quickly”
10	“I liked the specific instructions like “from the right lane” it made it feel more like a real GPS”
11	“Nothing really stood out as distracting”
12	-
13	“The new message alerts felt different from the nav ones. Took me an extra moment to adjust.”
14	“The simpler audio alerts were definitely the

	hardest for me to catch”
15	“No additional comments”
16	-
17	“Bright icons worked really well. I could react without really thinking”
18	-
19	“Audio was clear”
20	“It felt most like a real dashboard when the icons were bright”
21	“The pacing felt natural, some of the alerts took me by surprise which I think is the point”
22	-
23	“I definitely preferred the shorter ones”
24	“The bright icons are what I expect from a modern car”